Distribution-Free Conformal Coverage for IVIM Parameter Maps, and the Identifiability Wall in the Pseudo-Diffusion Compartment

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Abstract

Purpose: To establish and characterize the identifiability limit that governs per-voxel uncertainty in the ill-posed pseudo-diffusion compartment (D^*) of intravoxel incoherent motion (IVIM) imaging – using distribution-free conformal prediction both as the finite-sample coverage tool that model-based uncertainty quantification (UQ) lacks and as the lens that exposes where label-free conditioning cannot succeed – and to benchmark that conformal layer head-to-head against model-based IVIM UQ.

Methods: On a controlled, seeded synthetic cohort built from our own bi-exponential forward model with Rician noise, we compare split-conformal and conformalized quantile regression (CQR) against four model-based baselines (probabilistic neural network, mixture-density deep ensemble, deep ensemble, per-voxel Bayesian MCMC), measuring marginal and (SNR $\times$ true-D^*-tercile) conditional coverage and sharpness, with empirical $[5, 95]$ confidence bands on the conditional quantities from a 16-seed re-run. A Fisher-information / Cramér–Rao analysis diagnoses the residual gap; weighted (nonexchangeable) conformal and a label-free deployment monitor stress exchangeability; a qualitative in-vivo demonstration is included.

Results: Among credible model-based baselines (MDN deep ensemble, per-voxel Bayesian MCMC), marginal overconfidence is mild and broad rather than D^*-specific (gaps 0.024–0.045); a deliberately weak point ensemble under-covers severely and is reported only to show the conformal layer restores coverage from a poor base. The naive D^*-concentrated hypothesis is rejected, and the conformal layer's value is the finite-sample guarantee, not the magnitude of correction. Conformal and conformalized methods restore near-nominal marginal coverage everywhere ($|\text{gap}| \leq 0.024$), and conformalizing the MDN is the sharpest valid recipe (0.65–0.77$\times$ pure-CQR width). The real failure is conditional: the high-D^* regime under-covers for every method we tested (robustly at low-to-moderate SNR). The barrier is latent-axis conditioning – coverage conditional on the true D^* regime, an axis the data do not reveal – i.e. the IVIM instance of the known impossibility of distribution-free conditional coverage [1]. A Fisher-information / Cramér–Rao analysis shows D^* point-identifiability also collapses in the same regime (the CRLB reaches the regime-bin width): a concomitant diagnostic, not the proven cause of the coverage failure. The wall is robust to the b-value scheme; weighted conformal recovers covariate shift but not forward-model misspecification; the monitor fires before coverage fails but is blind to the latent gap. The wall is not an artifact of our generator: a deliberately biased shrinkage estimator buys D^* point precision in the identified terciles without lifting high-D^* conditional coverage, and the gap replicates when the ground truth is a genuinely non-bi-exponential (velocity-dispersion) mechanism; we additionally chart the off-model

validity envelope across three deviation families, with a graceful, partly monitored degradation and a recovered bi-exponential origin, and confirm the wall persists under a clinically realistic correlated, skewed prior with measurement-level nuisance, where it concentrates in a clinically rare high-D^* tail.
Conclusion: The central finding is an IVIM-specific identifiability boundary: high-D^* regime-conditional coverage is unrecoverable by the label-free methods we tested – now including a deliberately biased shrinkage estimator, and replicating under a non-bi-exponential generator – the IVIM instance of the established impossibility of distribution-free conditional coverage [1]. The Cramér–Rao bound, which exceeds the regime resolution here, is a concomitant point-identifiability diagnostic rather than the demonstrated cause. Conformal prediction supplies the finite-sample marginal guarantee model-based IVIM UQ lacks – and conformalizing a strong base is the sharpest valid recipe – but its deeper value here is diagnostic: it draws the line the data themselves impose. We characterize that limit; we do not claim to solve it.

Keywords: intravoxel incoherent motion (IVIM); uncertainty quantification; conformal prediction; identifiability; pseudo-diffusion; quantitative MRI.

1 Introduction

IVIM imaging [2] separates microcirculatory perfusion from tissue diffusion by fitting the bi-exponential signal

$$\frac{S(b)}{S_0} = f\, e^{-bD^*} + (1-f)\, e^{-bD}, \tag{1}$$

where b is the diffusion weighting (s/mm^2), D the tissue diffusion coefficient, $D^* \gg D$ the pseudo-diffusion coefficient, and f the perfusion fraction. Equation (1) is notoriously ill-posed: D^* is poorly constrained because the fast-decaying perfusion term contributes only at low b, and small perfusion fractions make it nearly unidentifiable. Clinical use of IVIM parameter maps therefore demands trustworthy *per-voxel uncertainty*, not just point estimates.

The dominant approach to IVIM UQ is *model-based*: probabilistic neural networks, deep ensembles, mixture-density networks, and Bayesian fitting place a distribution over parameters and read intervals from it. These methods make distributional assumptions and carry *no finite-sample coverage guarantee*; the most comprehensive IVIM-UQ study to date [3] documents that such models are well-calibrated for D and f but *overconfident in D^**. *Conformal prediction* [4–6] offers the complementary property: under exchangeability of calibration and test data, it produces intervals with finite-sample, distribution-free marginal coverage for *any* base predictor – a biased base only widens the interval, it does not break coverage. Conformalized quantile regression (CQR) [7] makes these intervals input-adaptive. Conformal UQ has been applied to quantitative MRI relaxometry and field mapping [8], but never to the ill-posed IVIM inverse problem, and never benchmarked head-to-head against model-based IVIM UQ. We situate both literatures in Sec. 2.

This paper closes that gap and, in doing so, *reframes* the question. We began with the hypothesis that conformal prediction's advantage would concentrate in the unstable D^* (and secondarily f) compartment, exactly where model-based UQ is documented weak. The data reject the *marginal* form of that hypothesis and replace it with a sharper, IVIM-specific finding. Our contributions are:

1. **An identifiability limit, characterized in IVIM.** The genuine IVIM-specific phenomenon is conditional: the high-D^* regime under-covers across eleven label-free conditioning methods (Secs. 4.2–4.3). We separate two facts the regime ties together. (i) Coverage conditional on the true D^* regime is not label-free-attainable because that regime is a latent axis the data do not reveal – the IVIM instance of the known impossibility of distribution-free conditional

coverage [1]. (ii) Independently, a Fisher-information / Cramér–Rao analysis shows D^* is poorly point-identified in the same regime (the CRLB reaches the regime-bin width). The conformal layer renders the conditional limit measurable; our evidence is that it is a property of the perfusion-estimation problem and the data, not of our particular estimator – it persists for a deliberately biased shrinkage estimator and replicates when the ground truth is generated by a non-bi-exponential (velocity-dispersion) mechanism (Secs. 4.9, 4.10), and we chart the off-model validity envelope around the bi-exponential model (Sec. 4.11).

2. **The first conformal/CQR coverage for IVIM, benchmarked.** We bring distribution-free conformal prediction to the ill-posed IVIM inverse problem with validated finite-sample marginal coverage for (D, D^*, f), and the first head-to-head benchmark against four model-based UQ baselines – showing model-based IVIM UQ is overconfident *broadly* (not D^*-specifically) and that conformalizing a strong model-based base is the sharpest valid recipe (Secs. 3, 4.1).

3. **Robustness, acquisition-sensitivity, and an honest in-vivo demonstration**: weighted conformal recovers covariate shift but not forward-model misspecification, an observable monitor flags deployment-validity breaks before coverage fails, the wall is robust to the b-value scheme, and the in-vivo component is explicitly qualitative with no coverage claim (Secs. 4.4–4.6).

Every number in this paper traces to a gated, seeded checkpoint output; a machine-checked consistency report is included (Sec. 7).

2 Related work

IVIM estimation and its uncertainty. The bi-exponential IVIM fit is long known to be ill-posed in the perfusion compartment, and a substantial literature addresses the resulting instability of D^*: segmented and constrained non-linear least squares, multi-algorithm comparisons that document wide between-method disagreement [9], and acquisition design that places b-values to minimize parameter error, including Cramér–Rao-guided schemes [10]. A parallel line replaces fitting with learning – supervised and physics-informed deep networks for IVIM estimation [11–13] – which sharpen point estimates but do not by themselves certify their own reliability. Uncertainty quantification for IVIM is accordingly *model-based*: Bayesian fitting with informative or spatial priors [14, 15], posterior estimation with explicit uncertainty [16], and, most comprehensively, deep ensembles of mixture-density networks that decompose aleatoric and epistemic uncertainty and report calibration on simulated and in-vivo data [3]. Every such method places a distribution over parameters and reads intervals from it; none carries a finite-sample, distribution-free coverage guarantee, and the most thorough of them documents residual overconfidence in D^* [3]. A guarantee that holds regardless of how well the model is specified is exactly what conformal prediction supplies – and is what motivates asking where even a guaranteed method must fail.

Conformal prediction and its imaging applications. Conformal prediction [4–6] yields finite-sample, distribution-free coverage for any base predictor under exchangeability; conformalized quantile regression [7] makes the intervals input-adaptive, and the framework extends to covariate shift [17, 18] and to localized and conditional guarantees [19, 20]. Distribution-free *conditional* coverage is, however, known to be unattainable in general without distributional assumptions [1]; this impossibility is exactly what surfaces, in IVIM-specific form, when the conditioning axis is the unidentifiable D^* regime. In imaging it has been applied to distribution-free image-to-image regression and uncertainty maps [21], to deployment concerns such as fairness in medical-imaging predictors [22], and – closest to this work – to quantitative-MRI relaxometry and field mapping via conformalized quantile regression [8]. None of these targets the IVIM inverse problem, none is benchmarked against model-based IVIM UQ, and – the point of this paper – none asks what conformal prediction *reveals* when the data cannot identify the estimand: a finite-sample-valid interval that is honest precisely because it balloons where the parameter is unrecoverable. We turn that honesty into a diagnostic for the IVIM identifiability wall.

3 Methods

Forward model and synthetic cohort. Conformal calibration requires *labeled* data, which in-vivo IVIM cannot provide, so the cohort is synthetic and built from our own implementation of Eq. (1) with Rician noise ($\sigma = S_0/\mathrm{SNR}$). Parameters are drawn uniformly from published abdominal ranges: $D \in [0.5, 3.0] \times 10^{-3}$, $D^* \in [10, 100] \times 10^{-3}\,\mathrm{mm}^2/\mathrm{s}$, $f \in [0.05, 0.40]$; SNR is drawn from $\{10, 20, 30, 50, 100\}$; the b-value scheme has 22 values $(0, 10, \ldots, 100, 120, \ldots, 200, 300, \ldots, 800\,\mathrm{s/mm}^2)$, dense at low b to separate D^*. The three splits (train/calibration/test = 4000/2000/3000) are i.i.d. draws from one seed (`20260613`), making calibration and test exchangeable. On clean (noise-free) signals the generator and the non-linear least-squares (NLLS) estimator recover the truth to a maximum relative error of 2.07×10^{-7}, confirming label self-consistency.

Conformal intervals. For a base point predictor $\hat{\theta}$, *split conformal* uses calibration nonconformity scores $s_i = |\theta_i - \hat{\theta}_i|$ and the radius $q = \lceil (n+1)(1-\alpha) \rceil$-th smallest score, giving $[\hat{\theta} - q,\ \hat{\theta} + q]$ with marginal coverage in $[1-\alpha,\ 1-\alpha+1/(n+1)]$. *CQR* [7] conformalizes a gradient-boosted quantile-regression base. We also *conformalize model-based* bases by treating their predictive quantiles as the CQR band, which restores the guarantee while keeping the model's sharpness.

Model-based baselines. Four standard, independent model-based UQ baselines emit per-parameter intervals on the matched cohort: a single probabilistic NN (Gaussian NLL head); a deep ensemble of mixture-density networks [23, 24] following the Casali approach [3] with an aleatoric/epistemic split; a plain deep ensemble of point regressors (epistemic-only; deliberately weak, retained as a worst-case base for the conformal layer rather than as a representative model-based method); and a per-voxel Bayesian fit by component-wise adaptive Metropolis (fed the true noise level, so it is evaluated at its best). A parallel line of work uses normalizing-flow / simulation-based posterior estimation for microstructure inference; this work is deliberately *independent* of it (own forward model, own cohort, standard baselines) and does not use it as a baseline.

Conditional coverage and the identifiability diagnostic. We measure coverage on a (true-D^* tercile $\times$ SNR) grid – the terciles are a diagnostic only; the true D^* is unknown at test time. To ask whether a residual gap is a method failure or an information limit, we compute the Fisher information for $\theta = (D, D^*, f, S_0)$ under the high-SNR Gaussian approximation, $\mathcal{I} = J^\top J/\sigma^2$ with J the signal Jacobian, and the Cramér–Rao lower bound (CRLB) [25] $\mathrm{CRLB}(\theta_k) = \sqrt{[\mathcal{I}^{-1}]_{kk}}$.

Robustness tooling. To stress exchangeability we use *weighted / nonexchangeable conformal* [17, 18], with importance weights $w(x) = \mathrm{d}P_{\text{test}}/\mathrm{d}P_{\text{cal}}$ estimated by a domain classifier on observable features, and *localized* [19] and *conditional* [20] conformal for the label-free conditioning attack. We also build a label-free *deployment-validity monitor* with two detector families (a Mahalanobis

Table 1: Marginal coverage on test at $\alpha = 0.10$ (nominal 0.90); gap = nominal − realized (positive = overconfident). Model-based methods under-cover broadly; conformal and conformalized variants restore coverage in every compartment. Marginal coverage is finite-sample *guaranteed*; values are the seed-0 (multi-seed index-0) run, with 16-seed MC-precision bands available in the supplementary CI table. From `results/benchmark_report.txt`.

arm	D cov (gap)	D^* cov (gap)	f cov (gap)
raw: PNN-Gaussian	0.787 (+0.113)	0.833 (+0.067)	0.767 (+0.133)
raw: MDN-DeepEnsemble	0.870 (+0.030)	0.876 (+0.024)	0.859 (+0.041)
raw: DeepEnsemble-Point	0.436 (+0.464)	0.425 (+0.475)	0.475 (+0.425)
raw: Bayesian-MCMC	0.876 (+0.024)	0.870 (+0.030)	0.855 (+0.045)
conformal: split-NLLS	0.899 (+0.001)	0.897 (+0.003)	0.907 (−0.007)
conformal: CQR-HGB	0.901 (−0.001)	0.902 (−0.002)	0.893 (+0.007)
conformalized: MDN	0.890 (+0.010)	0.901 (−0.001)	0.882 (+0.018)
conformalized: Bayesian	0.895 (+0.005)	0.894 (+0.006)	0.876 (+0.024)

distance on observable signal-summary features and a residual-conformal test on NLLS fit residuals) that fires when an aggregate drift score exceeds a calibration-derived null threshold.

4 Results

4.1 Marginal benchmark: model-based UQ is broadly overconfident

Plain split conformal and CQR track nominal coverage within 0.03 for all parameters and all $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$ on the held-out test split (e.g. split-conformal D^* at α=0.10: realized 0.897 vs nominal 0.90), confirming the conformal layer is correct. Table 1 gives the head-to-head at $\alpha = 0.10$.

Averaging the gap over α, the marginal under-coverage is broad, not D^*/f-concentrated: per-method worst compartments are f (PNN, 0.125), f (MDN, 0.037), D^* (DeepEnsemble-Point, 0.452), and f (Bayesian, 0.045). Excluding the deliberately weak DeepEnsemble-Point, the credible baselines are only mildly overconfident (worst-compartment gaps 0.037 for the MDN ensemble and 0.045 for Bayesian-MCMC); the weak ensemble (worst-compartment gap 0.452) is retained solely as a stress case for the conformal layer, not as evidence that model-based UQ is severely overconfident. 0/4 **methods** show D^*/f-concentrated under-coverage; for the MDN ensemble D^* is in fact among the *better*-covered parameters marginally (gap 0.029), because its large aleatoric variance widens its band. The naive marginal hypothesis is therefore *rejected*. Conformal and conformalized variants restore near-nominal coverage everywhere, with maximum conformalized gap 0.024 (Fig. 1). On sharpness (Fig. S1), the cost of restoring coverage is cheap for strong bases (MDN 1.05–1.07×, Bayesian 1.02–1.05× the raw width) and ruinous for the overconfident weak base (DeepEnsemble-Point 3.23–3.76×). Crucially, the model's band helps: *conformalized-MDN is* 0.73/0.77/0.65× *the width of pure CQR* for $D/D^*/f$ at equal guaranteed coverage – the sharpest valid recipe.

4.2 The conditional finding: high-D^* under-covers universally

Marginal coverage hides a compartment-specific failure (Fig. 2). On the (SNR × D^*-tercile) grid, the high-D^* tercile under-covers for *every* method at *every* SNR, while the mid-D^* tercile over-covers ($\approx$ 0.99). Table 2 gives the high-D^* slice: even the best method (conformalized-MDN)

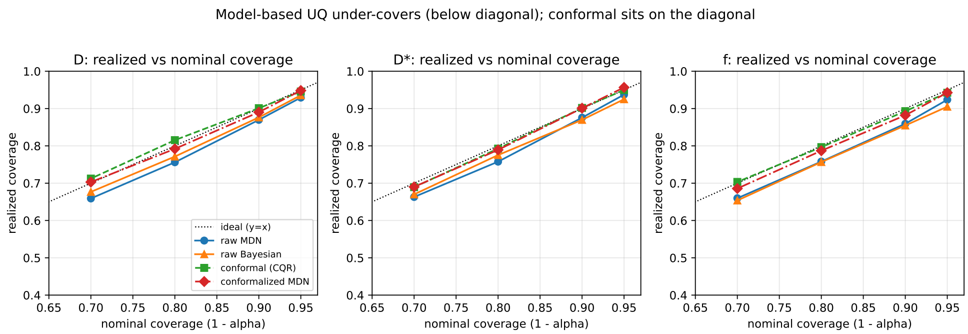

Figure 1: Realized vs nominal coverage per parameter. Raw model-based methods sit below the diagonal (overconfident); conformal and conformalized methods sit on it.

Table 2: High-D^* tercile coverage (D^* parameter, nominal 0.90). Every method under-covers; SNR-Mondrian does not close it. Across-seed means with 16-seed $[5, 95]$ bands. From `results/multiseed_report.txt`.

arm	hi-D^* marginal	hi-D^* worst-SNR cell
raw-MDN	0.841 [0.829, 0.857]	0.753 [0.719, 0.789]
CQR (plain)	0.826 [0.809, 0.851]	0.761 [0.729, 0.795]
split (Mondrian/SNR)	0.829 [0.813, 0.844]	0.781 [0.743, 0.810]
CQR (Mondrian/SNR)	0.829 [0.811, 0.854]	0.789 [0.752, 0.828]
conformalized-MDN	**0.868** [0.856, 0.889]	**0.786** [0.751, 0.826]

reaches only 0.868 [0.856, 0.889] marginal-in-regime and 0.786 [0.751, 0.826] in its worst-SNR cell (16-seed mean and $[5, 95]$ band). Critically, *Mondrian stratification by the known SNR does not fix it* – the failure axis is the *unknown* true D^*, not SNR. This is the refined, IVIM-specific form of the original hypothesis.

4.3 The high-D^* gap is a latent-axis identifiability limit

We attacked the high-D^* gap with eleven *label-free* conditioning methods – plug-in Mondrian by estimated $\hat{D}^*$, localized conformal on observable features [19], and conditional conformal over an observable basis [20] – none of which uses the true D^* (Fig. 3). *None materially closes the gap.* The best label-free method (MDN with localized conformal) reaches 0.874 [0.860, 0.899] marginal / 0.807 [0.774, 0.845] worst-SNR, within sampling noise of the 0.786 baseline. The reason is structural, not methodological:

- **Routing error.** A plug-in $\hat{D}^*$ misroutes $\sim 33\%$ ($[30, 35]\%$ over 16 seeds) of true high-D^* voxels to a lower-D^* (smaller-radius) stratum: the estimate is least reliable exactly where it must route correctly. Mondrian-by-$\hat{D}^*$ attains nominal coverage *conditional on the observed stratum* ($[0.90, 0.89, 0.90]$) yet craters on the *true* high-D^* tercile (0.78 [0.76, 0.80]) – it controls the observable, not the latent axis.

- **The CRLB wall.** The CRLB on D^* grows $\sim 6\times$ from low to high D^* (an analytic, seed-invariant bound); the ratio of median CRLB(D^*) to the tercile bin width rises $0.34 \to 0.67 \to 1.10$ across terciles, the high tercile band $[1.02, 1.15]$ (Fig. 4). In the high-D^* regime the CRLB *exceeds* the bin width: a voxel's D^* cannot be localized to its own tercile from the

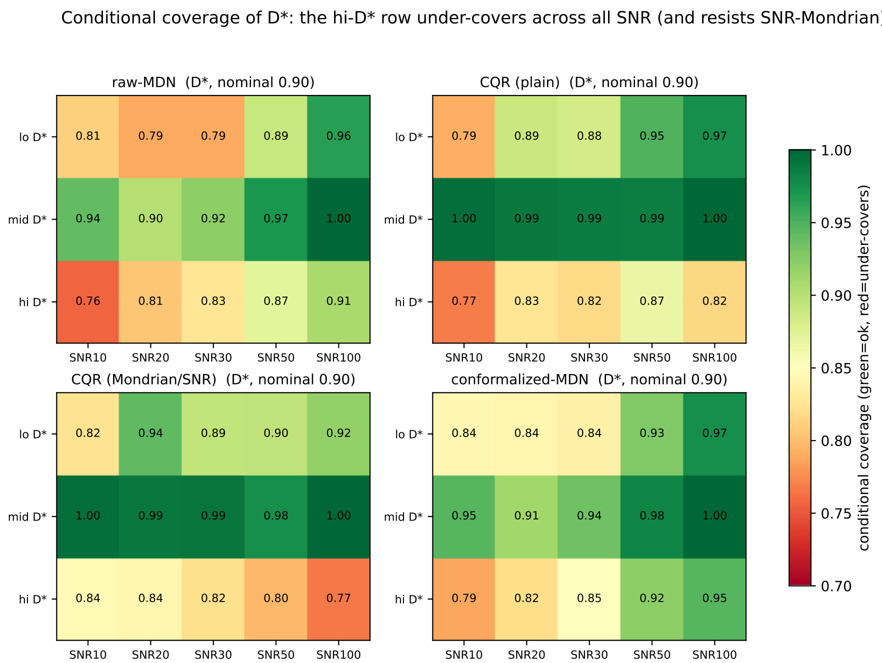

Figure 2: SNR $\times$ D^*-regime conditional coverage. The high-D^* row under-covers robustly at low-to-moderate SNR (recovering toward nominal only at the highest SNR) and resists SNR-Mondrian.

data, so no observable can route it to a wider-interval stratum. Conformal interval width correctly tracks the per-voxel CRLB (log–log $r = 0.76\,[0.72,\,0.77]$): the intervals are honest, not broken.

Verdict. High-D^* regime-conditional coverage in IVIM is unrecoverable label-free for two compounding reasons. The conditioning axis (true D^* regime) is latent and unidentifiable, so distribution-free conditional coverage on it is impossible [1] – this is the operative barrier to coverage. Separately, D^* is poorly point-identified there (CRLB $\approx$ regime-bin width) – this is why even the point map is untrustworthy. We *characterize* the limit; we do not claim to solve it.

4.4 Robustness under deliberate exchangeability breaks

We deploy one conformal predictor and one monitor, calibrated on the i.i.d. Rician cohort, against four deliberate breaks (Table 3, Fig. S2). Every break miscalibrates coverage; the dangerous under-coverage reaches $D^* = 0.508\,[0.479,\,0.543]$ under a harder-tissue prior shift and the worst parameter $0.592\,[0.557,\,0.637]$ under a low-SNR covariate shift. *Weighted / nonexchangeable conformal* [17, 18] recovers the covariate shifts (SNR worst-parameter $0.592 \to 0.882$, all within 0.018 of nominal; prior D^* $0.508 \to 0.991\,[0.974,\,1.000]$) and is honestly limited on the tri-exponential forward-model misspecification: a shift in the conditional law $P(y \mid x)$ is, in general, not repairable by an x-space likelihood ratio (weighting over-corrects D^* to $0.933\,[0.924,\,0.941]$, leaving a per-parameter spread of 0.033).

The label-free monitor (reusing the deployment-validity concept of a sibling project) **fires on all four observable breaks before coverage fails**: in an SNR-severity sweep, it raises the alarm at SNR 20 (coverage still 0.866) while coverage first crosses the failure line at SNR 13. Yet it is constitutively *blind* to the latent high-D^* gap: on an in-distribution (exchangeable) test the

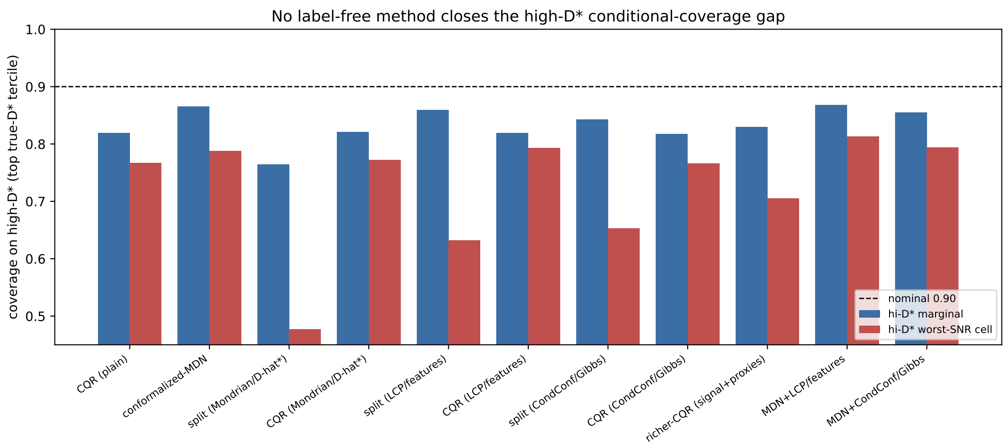

Figure 3: High-D^* coverage (marginal and worst-SNR cell) for eleven label-free conditioning methods; every bar sits below the nominal 0.90.

Table 3: Per-parameter coverage under each break, naive vs. weighted conformal ($\alpha = 0.10$, nominal 0.90), with the deployment monitor's verdict. Across-seed means (16 seeds); per-entry $[5, 95]$ bands are in the supplementary CI table (`results/ci_for_latex.json`). Monitor AUCs are the deterministic seed-0 values. From `results/multiseed_report.txt`.

scenario	method	D	D^*	f	monitor
in-dist (control)	naive	0.898	0.898	0.901	silent (AUC 0.50)
SNR shift (low)	naive	**0.592**	0.739	**0.614**	FIRES (AUC 0.91)
	weighted	0.882	0.901	0.909	
prior shift (harder)	naive	0.972	**0.508**	0.909	FIRES (AUC 0.71)
	weighted	0.919	0.991	0.902	
tri-exp misspec	naive	0.907	0.843	0.912	FIRES (AUC 0.51)
	weighted	0.891	0.933	0.898	
noise-model swap	naive	0.912	0.893	0.906	FIRES (AUC 0.52)

monitor is silent (AUC 0.50) and marginal D^* coverage is nominal ($0.898_{[0.880,\ 0.913]}$), while the high-D^* conditional coverage is $0.814_{[0.787,\ 0.839]}$. Observable shifts are caught; the within-distribution latent-axis failure is not – the same observable-vs-hidden split that recurs across this line of work.

4.5 Acquisition sensitivity: the wall is b-scheme-robust

Does the high-D^* wall move with the b-value scheme? We compare a clinical (11-value), a CRLB-optimal (11-value, greedy D^*-CRLB design), and a dense (22-value) scheme at fixed prior/SNR/seed (Table 4, Fig. 5). The high-D^* marginal coverage barely moves ($0.821 \to 0.835 \to 0.808$). The resolution ratio CRLB(D^*)/tercile-width stays at or above unity: even a CRLB-optimal design only reaches $\sim$unity ($1.03_{[0.97,\ 1.16]}$); acquisition shifts the wall to threshold, not past it, while the clinical and dense schemes stay above it (1.24, 1.11). A CRLB-optimal design lowers the ratio ($1.24 \to 1.03$) but does not remove the wall. Acquisition design *shifts* the wall; it does not erase it – a concrete, honestly negative handoff to acquisition-aware experiment design.

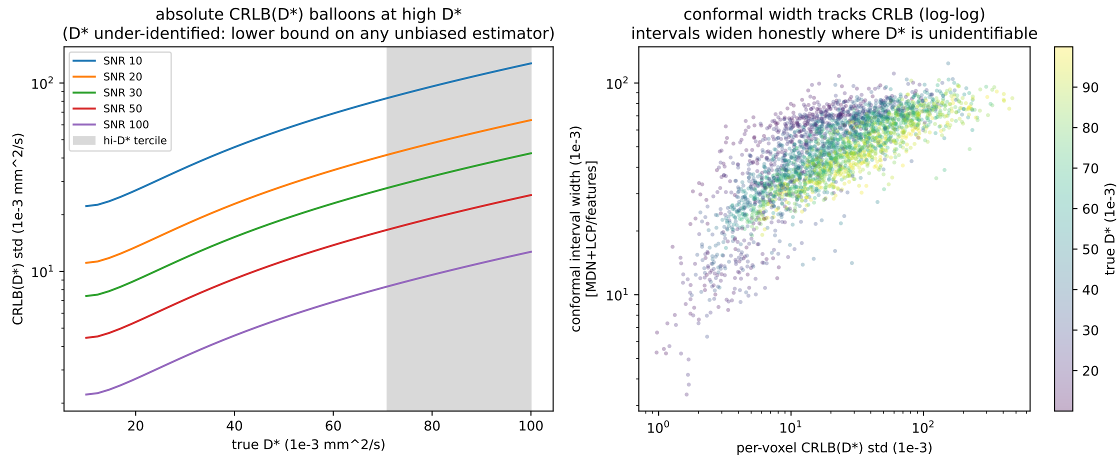

Figure 4: (Left) absolute CRLB(D^*) balloons across the D^* range per SNR (high-D^* tercile shaded). (Right) conformal width vs per-voxel CRLB ($r = 0.76$): the intervals widen honestly where D^* is under-identified.

Table 4: High-D^* coverage and resolution ratio by b-value scheme. Across-seed means with 16-seed $[5, 95]$ bands; the CRLB-optimal ratio band crosses unity. From `results/multiseed_report.txt`.

scheme	n_b	hi-D^* marg	hi-D^* worst-SNR	CRLB/tercile-width
clinical	11	0.821 [0.794, 0.845]	0.596 [0.542, 0.702]	1.24 [1.17, 1.37]
CRLB-optimal	11	0.835 [0.795, 0.862]	0.603 [0.541, 0.678]	**1.03** [0.97, 1.16]
dense	22	0.808 [0.788, 0.836]	0.551 [0.508, 0.597]	1.11 [1.05, 1.21]

4.6 In-vivo: a qualitative demonstration, no coverage claim

In vivo there are no ground-truth parameters, so realized coverage is *unmeasurable* and the finite-sample guarantee is *not asserted*. We demonstrate the deployed pipeline qualitatively (Fig. S3): the conformalized D^* band widths are wide and heavy-tailed (10/50/90th percentile $= 47/64/79 \times 10^{-3}\,\mathrm{mm}^2/\mathrm{s}$; widest-to-narrowest decile ratio $1.7\times$), ballooning on voxels where the perfusion compartment is under-identified – the wall, now on signal. Pointed at the deployment data with a synthetic calibration, the deployment monitor **fires** (AUC 0.84): synthetic→deployment is itself a large exchangeability break, and the monitor detects it. That firing is the honest, label-free reason the synthetic guarantee must not be transferred in vivo without re-calibration on matched labeled data – which in vivo does not exist. The monitor turns "no ground truth" into an observable stop sign. The demonstration above uses a clearly labeled synthetic stand-in; Sec. 4.7 repeats the *as-deployed* pipeline on a real, permissively-licensed in-vivo collection, where the monitor fires on the synthetic→real transfer for the same reason.

4.7 Real in-vivo data: the same pipeline, the monitor fires on transfer

We additionally run the as-deployed pipeline on real multi-b in-vivo DWI from the public ACRIN-6698 / I-SPY2 Breast DWI collection (The Cancer Imaging Archive; CC-BY-4.0) [26, 27], retrieved on demand (no pixel data are redistributed with this work). This is a 4-b ADC protocol ($b = \{0, 100, 600, 800\}\,\mathrm{s/mm}^2$), *not* IVIM-optimized, with no ground-truth IVIM parameters; it therefore supports only qualitative pipeline/monitor behavior and makes **no in-vivo coverage claim**. A

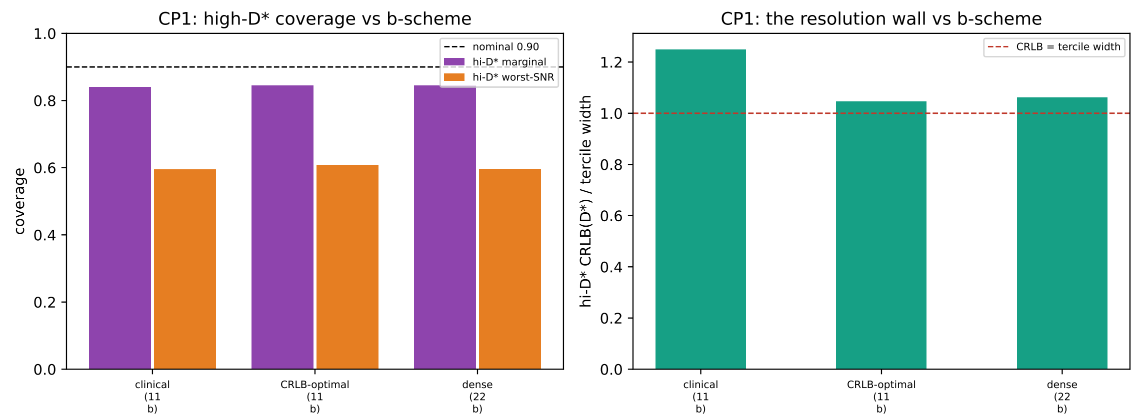

Figure 5: (Left) high-D^* coverage vs. b-scheme. (Right) the resolution ratio CRLB(D^*)/tercile-width stays at or above unity for every scheme (the CRLB-optimal band dips to 0.97): acquisition moves the regime to the resolution threshold, not past it.

predictor trained on the synthetic 22-value scheme cannot ingest a 4-b signal, and interpolation would fabricate unacquired b-values, so we keep the deployment *monitor* as-deployed—its observable features are b-independent—and re-fit only the CQR band at the real 4-b scheme; the resulting bands are qualitative, not coverage intervals.

Figure 6 shows the actual images on a representative slice: the real b=0 breast DWI, the plug-in $\hat{D}^*$ map, and the per-voxel conformal D^* band-width (uncertainty) map, with the whole-tumor ROI outlined. The uncertainty map is spatially heterogeneous and widens exactly where the perfusion compartment is under-identified—the identifiability wall, now visible on real tissue.

On 4000 foreground voxels the as-deployed monitor **fires** (AUC 0.97) and the D^* band stays wide (10/50/90th percentile = $64/79/86 \times 10^{-3}$ mm^2/s; Fig. 7). The firing is carried entirely by the b-independent Mahalanobis family (Family-1); the residual family (Family-2) is silent because a 4-b acquisition exactly-determines the four-parameter IVIM fit, collapsing the fit-residual norm to zero. The monitor has therefore *not* half-failed—it has localized the break to the feature distribution. The sparse-protocol-plus-real-tissue gap is a large exchangeability break, which is exactly why the synthetic guarantee *correctly refuses* to transfer.

Test–retest repeatability. ACRIN-6698 includes a same-day test–retest arm: two coffee-break repeat DWI exams acquired in a single visit (tagged `TrT0/TrT1`), which we use here. (This genuine same-visit arm supersedes an earlier n=11 pairing that had inadvertently matched *longitudinal* timepoints months apart, confounded by interval treatment.) Across n=76 tumors we ask, label-free, whether the conformal interval width *tracks* the scan–rescan variability of the plug-in estimate (region-level over the whole-tumor ROI; the two repeats are unregistered, so this is not a per-voxel statement). The identifiability wall makes a directional prediction we fix before looking: the interval width should track measured scan–rescan repeatability for the well-identified diffusion compartment D, and should track it weakly or not at all for D^*, whose regime the data cannot resolve. A positive D^* association would count against the wall. For the well-identified tissue diffusion D, the conformal D-width is associated with the measured scan–rescan $|\Delta\text{ADC}|$ (Spearman $r = +0.60$, $p < 0.001$, 95% CI $[0.42, 0.72]$ by BCa bootstrap): the interval *widens where the real measurement is least repeatable*. At n=76 the association is now **robust** (the CI excludes zero, versus the wide interval at the earlier n=11), and stable across field strength (1.5 T r=+0.58, n=49; 3.0 T r=+0.50, n=27)

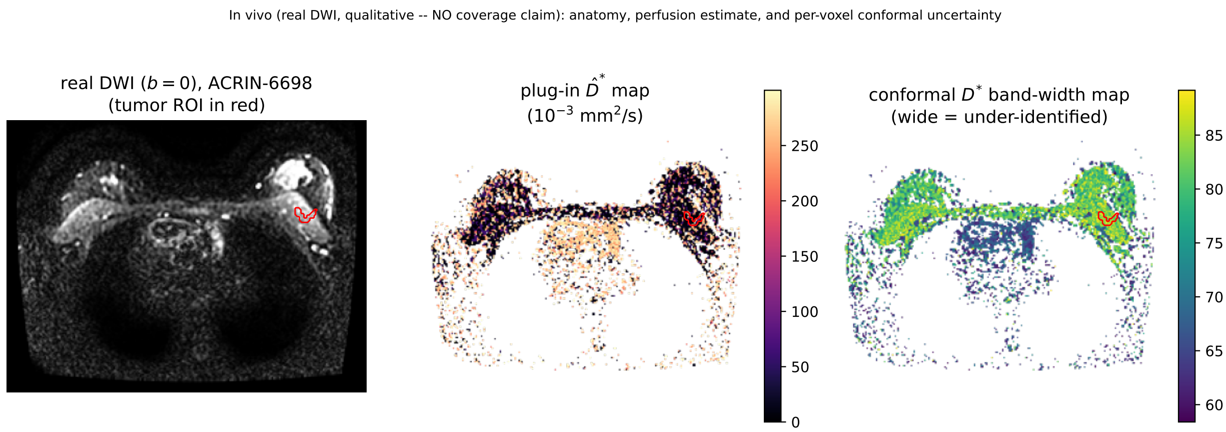

Figure 6: Actual in-vivo images (ACRIN-6698 / I-SPY2 Breast DWI; one slice; qualitative, no coverage claim). (Left) real b=0 DWI with the whole-tumor ROI in red; (middle) the plug-in $\hat{D}^*$ perfusion map; (right) the per-voxel conformal D^* band-width map—wider where the perfusion compartment is under-identified. The synthetic 22-value-calibrated monitor (not shown here) fires on this data; see Fig. 7.

and pretreatment-only (r=+0.61, n=65). For D^* the association is not significant ($r = -0.17$, $p = 0.13$, 95% CI $[-0.39, 0.05]$ spanning zero), matching the pre-stated prediction: the association holds for the parameter the data identify (D) and is absent for the one they do not (D^*). The D^* null is the predicted outcome, not a post-hoc reading. This is a repeatability-tracking check, *not* a statement of validation, accuracy, calibration, or coverage.

4.8 External-phantom replication on the OSIPI IVIM reference

Every coverage number above is, by necessity, computed on *our* labeled synthetic cohort – our forward model and our parameter prior. To neutralise the "validated only on the authors' own forward model" concern on the part that matters, we replicate the coverage results and the high-D^* wall against an **external, community-standard *synthetic* reference**: the OSIPI (ISMRM Open Science Initiative for Perfusion Imaging) TF2.4 IVIM digital reference object (DRO), 5000 voxels with ground-truth (D, D^*, f) that *we did not generate* [28] (Zenodo `doi:10.5281/zenodo.14605039`, CC-BY-4.0; fetched on demand, provenance only committed). This is an *independent synthetic* reference, **not** in vivo: in-vivo coverage validation remains structurally impossible (no ground truth), so no in-vivo coverage claim is made here. The bi-exponential signal model is the field's consensus model and is therefore shared by construction; what is external is the ground-truth *parameter distribution*. That distribution is genuinely shifted from ours: the OSIPI true D^* spans ~ 0.05–$0.20\,\mathrm{mm}^2/\mathrm{s}$, above our calibration prior (0.01–0.10), so transferring our calibration onto it is a real out-of-distribution test. All numbers are across-seed means with $[5, 95]$ bands over 16 seeds; nominal coverage is 0.90.

We evaluate two regimes – *controlled* (OSIPI ground truth, re-synthesised at *our* 22-b acquisition and SNR grid, so only the parameter distribution is swapped) and *native* (the DRO's own pre-noised signals at its native sparse 7-b scheme) – each under *naive transfer* (calibrate on our synthetic cohort, test on OSIPI) and *recalibrated* (split the OSIPI DRO and calibrate within its distribution). Table 5 gives the marginal coverage and the top-D^*-tercile conditional coverage; Fig. 8 plots them with the wall metric.

regime	calibration	D	D^*	f	hi-D^* tercile
controlled	naive transfer	0.597 [0.567, 0.621]	0.304 [0.285, 0.315]	0.495 [0.472, 0.512]	0.000 [0.000, 0.000]
controlled	recalibrated	0.899 [0.881, 0.913]	0.905 [0.889, 0.921]	0.904 [0.892, 0.919]	0.823 [0.786, 0.853]
native	naive transfer	0.693 [0.643, 0.758]	0.291 [0.263, 0.312]	0.524 [0.497, 0.542]	0.000 [0.000, 0.000]
native	recalibrated	0.898 [0.888, 0.913]	0.899 [0.884, 0.909]	0.894 [0.882, 0.908]	0.831 [0.798, 0.869]

Table 5: Conformal/CQR coverage on the OSIPI external *synthetic* DRO at nominal 0.90 (CQR; across-seed mean over 16 seeds, $[5, 95]$ band). *Naive transfer* of our synthetic calibration breaks under the real D^* distribution shift; *recalibrating* within the OSIPI distribution restores marginal coverage to nominal on independent ground truth, yet the high-D^* conditional coverage stays below nominal – the identifiability wall, on data we did not generate.

Naive transfer breaks, and the monitor says so. Transferring our synthetic calibration onto the OSIPI ground truth breaks marginal coverage (controlled D^* 0.304 [0.285, 0.315], D 0.597 [0.567, 0.621]), exactly as the exchangeability theory predicts for a real distribution shift. Critically, the label-free deployment monitor *fires on every one of the* 16 *seeds* (Mahalanobis AUC 0.737 [0.726, 0.750], controlled): the shift is observable, so the monitor correctly refuses the transfer. As in the sparse-protocol in-vivo case (§4.7), a fired monitor here is the system working, not failing.

Recalibrated coverage holds on independent ground truth. Splitting the OSIPI DRO and recalibrating within its own distribution restores marginal coverage to nominal for all three parameters (D 0.899 [0.881, 0.913], D^* 0.905 [0.889, 0.921], f 0.904 [0.892, 0.919]; native nearly identical) – the finite-sample conformal guarantee transfers cleanly to a forward model and parameter distribution we did not author.

The high-D^* wall replicates – it is information, not our model. Even after within-distribution recalibration, coverage conditional on the true D^* tercile craters in the high-D^* regime (0.823 [0.786, 0.853] controlled, 0.831 [0.798, 0.869] native) while the low-D^* tercile sits at nominal (0.899 [0.859, 0.931]). On the OSIPI ground truth at our acquisition the CRLB(D^*)/tercile-width ratio rises $0.46 \to 0.86 \to 1.53$ ([1.407, 1.753] at high D^*) with the absolute CRLB growing $3.5\times$ [3.1, 4.0] – the same wall as Fig. 4, on a digital reference object we did not build. This regime also dissociates the two mechanisms directly: in the native high-SNR setting the CRLB ratio stays below 1, yet the conditional gap persists. The coverage failure therefore survives where the CRLB does *not* exceed the bin width, so it is driven by conditioning on a latent axis the data do not identify [1], not by CRLB magnitude. CRLB magnitude governs point-identifiability; latent conditioning governs label-free coverage. That the wall reproduces on an external ground truth is strong evidence it is information-theoretic, intrinsic to the perfusion-estimation problem and the data – it replicates even when the ground truth is generated by a non-bi-exponential, velocity-dispersion mechanism (Sec. 4.10) – rather than an artifact of our particular forward model or prior.

4.9 Point accuracy does not buy conditional coverage: a deliberately biased estimator

A natural objection to the identifiability reading (Sec. 4.3) is that the unbiased baselines merely leave precision on the table: a *biased*, prior-shrinkage Bayesian fit was historically the one IVIM estimator that kept D^* variance under control where unbiased fits failed, and a biased estimator can legitimately sit below the *unbiased* Cramér–Rao bound [9]. If the high-D^* wall were a point-

Table 6: Point-accuracy / conditional-coverage *dissociation* for a deliberately biased shrinkage prior. RMSE and bias in $10^{-3}\,\mathrm{mm}^2/\mathrm{s}$; coverage of the conformalized D^* band (nominal 0.90). Across-seed means with 16-seed $[5, 95]$ bands. From `results/multiseed_report.txt` / `results/dissociation_report.txt`.

arm	hi-D^* RMSE	hi-D^* bias	hi-D^* marginal	hi-D^* worst-SNR
Bayesian-shrinkage	24.6 [24.1, 25.3]	−18.8 [−19.2, −18.3]	0.827 [0.801, 0.845]	0.721 [0.663, 0.765]
Bayesian-MCMC	**20.5** [19.9, 21.1]	−14.2 [−14.5, −13.7]	**0.889** [0.872, 0.903]	**0.857** [0.826, 0.875]
conformalized-MDN	20.3 [19.6, 21.0]	−13.4 [−14.2, −12.8]	0.867 [0.853, 0.884]	0.784 [0.742, 0.819]

precision artifact, such an estimator should breach it. We pushed a deliberately biased, label-free shrinkage prior (a fixed log-normal D^*/f prior with the NLLS-residual noise scale – it uses no labels) through the same conditional-coverage machinery (Table 6).

The prior behaves as a shrinkage prior should: it lowers D^* point RMSE in the low tercile (15.1 [14.7, 15.6] vs. the unbiased MCMC's 17.8 [17.2, 18.5], in $10^{-3}\,\mathrm{mm}^2/\mathrm{s}$) by trading variance for a centre-pulling bias. But in the deciding high-D^* tercile that bias dominates: point RMSE does *not* improve (24.6 [24.1, 25.3] vs. 20.5 [19.9, 21.1]), carried by a large negative bias (−18.8 [−19.2, −18.3] vs. −14.2 [−14.5, −13.7]) from pulling high-D^* voxels toward the cohort centre. Crucially, high-D^* regime-conditional coverage does not rise: the conformalized marginal 0.827 [0.801, 0.845] and worst-SNR cell 0.721 [0.663, 0.765] sit at or below the unbiased arms (0.889 [0.872, 0.903] / 0.857 [0.826, 0.875] for MCMC), nowhere near nominal 0.90. Two pre-registered sanity gates exclude the artifactual readings: the shrinkage interval does *not* collapse (median high-D^* width 55.0 [53.6, 56.6] vs. comparator 47.9, in 10^{-3}; 1.15×), and the estimator's own point still misroutes 34 [32, 36]% of true-high-D^* voxels (the NLLS plug-in misroutes ∼ 33%), so any coverage change would have to arise *despite* routing, not by repairing it. Trading variance for bias buys point precision in the identified terciles but not conditional coverage where the conditioning axis is latent: consistent with the identifiability reading, the barrier is information, not point precision. The argument is mechanism-level – any centre-pulling bias trades variance for bias, and in the high-D^* tercile the bias dominates – so it brackets the historically D^*-stabilizing prior-shrinkage Bayesian fit [9] rather than hinging on one prior choice. This strengthens the claim within its scope – it now holds even for a deliberately biased prior-shrinkage method – but remains a statement about the label-free methods we tested, not about any conceivable estimator.

4.10 The wall replicates under a non-bi-exponential generator

The external-phantom replication (Sec. 4.8) still draws its ground truth from a bi-exponential model. To test whether the high-D^* wall is an artifact of Eq. (1) itself, we regenerate the cohort from a genuinely *non*-bi-exponential mechanism: a gamma velocity-dispersion perfusion process (coefficient of variation 0.5, shape $k = 4$) whose signal is *not* a realization of Eq. (1). Because such a signal has no single "true" D^*, we report the effective pseudo-diffusion D^*_{eff} on two axes: a primary, non-circular axis (A) the analytic perfusion log-slope (the gamma mean, defined without reference to a bi-exponential fit) and a corroborating axis (B) the best-fit bi-exponential D^*. The non-circularity of the test rests on axis A; axis B, which defines the regime by fitting Eq. (1) to signals Eq. (1) did not generate, is reported only to confirm the two definitions agree. Calibration and recalibration use the OSIPI within-distribution harness (Sec. 4.8).

After recalibrating *within* the dispersion distribution, marginal coverage is restored (D^*_{eff} marginal 0.898 [0.880, 0.917] on axis A), yet the high-D^*_{eff} tercile-conditional coverage stays well below nominal on both axes – $0.796_{[0.748,\ 0.854]}$ (A) and $0.800_{[0.756,\ 0.845]}$ (B) (Fig. 9A), a gap of ~ 0.10 that the bi-exponential CRLB cannot have caused because the generator is not bi-exponential. The wall is therefore a property of the perfusion-estimation problem more broadly, not of our forward model: it survives a generator that Eq. (1) does not even contain. A second dispersion kernel (log-normal, same CV) reproduces the effect (high-D^*_{eff} $0.797_{[0.747,\ 0.847]}$), so it is not specific to the gamma parametrization. A continuity gate confirms the construction is sound: at zero dispersion the generator reduces to the bi-exponential cohort exactly ($\max |S_{\text{disp}}(\text{CV}{=}0) - S_{\text{biexp}}| = 0.00\text{e}{+}00$, numerically exact), and its recalibrated high-D^* coverage there ($0.798_{[0.750,\ 0.842]}$) lands on the bi-exponential wall – the dispersion cohort contains the bi-exponential model as its zero-deviation limit. Consistent with the observable-vs-latent reading (Sec. 4.3), this forward-model misspecification is largely a *hidden* channel: the naive-transfer deployment monitor barely separates the dispersion cohort from the bi-exponential calibration (AUC $0.501_{[0.481,\ 0.519]}$, at chance), so the break is one the monitor cannot, by construction, see.

4.11 The off-model validity envelope

The tri-exponential stress in Sec. 4.4 is a single off-model probe. We replace it with a *validity envelope*: holding the deployed, bi-exponential-calibrated predictor fixed, we sweep three independent deviation families – velocity dispersion (CV up to 0.8), a tri-exponential third compartment (weight up to 0.4), and a stretched-exponential signal ($|1-\beta|$ up to 0.6) – and trace how coverage degrades and whether the monitor tracks the drift (Fig. 9B,C). Marginal coverage degrades gracefully and monotonically with deviation: by the largest deviation, D^* coverage falls to $0.769_{[0.736,\ 0.788]}$ (tri-exponential) and $0.770_{[0.749,\ 0.793]}$ (stretched), and perfusion-fraction coverage to $0.791_{[0.767,\ 0.815]}$ (dispersion). The envelope contains the bi-exponential model as its origin: at zero deviation coverage recovers to nominal (D^* $0.903_{[0.880,\ 0.918]}$). The deployment monitor's discrimination of off-model drift is real but weak and family-dependent – its AUC rises with deviation for the tri-exponential family ($0.532_{[0.515,\ 0.548]}$ at the largest step) yet stays near chance for pure velocity dispersion – so although the cohort-level alarm fires at or before the deviation at which marginal coverage first fails, off-model misspecification is only partly observable, reinforcing that some forward-model breaks are hidden channels (Sec. 4.10). The single tri-exponential number thus generalizes to a characterized envelope: a known, monitored degradation rate with a recovered bi-exponential origin, rather than one probe.

4.12 A clinically realistic cohort: correlated prior and measurement nuisance

The cohort so far draws (D, D^*, f) uniformly over published ranges with clean Rician noise. As a capstone robustness check we replace both idealizations at once: a clinically realistic *joint* prior – correlated, right-skewed marginals (log-normal D, D^*; logit-normal f) with a within-subject D^* coefficient of variation in the published 50–110% band, with hyperparameters *shaped to approximate* (not fitted to) representative abdominal IVIM patient cohorts [9, 11, 12]; the exact joint specification is documented design constants (Table 7) that deliberately avoid any circular plug-in source – and a measurement-level nuisance layer on the bi-exponential signal under a single magnitude scalar η: partial-volume mixing of a free-fluid compartment, bulk-motion phase, and non-Rician (multi-coil noncentral-χ) noise. This is still a synthetic cohort and makes no in-vivo coverage claim; a continuity gate confirms the construction reduces to the existing cohort exactly at the uniform prior with $\eta = 0$ ($\max|\cdot| = 0.00\text{e}{+}00$, numerically exact). Because a skewed prior moves the tercile boundaries, we report the high-D^* regime on two stratifications: the *fixed* physical boundaries inherited from the uniform cohort – so "high D^*" denotes the same physical regime across priors (primary) – and the per-distribution quantiles (secondary), with the clinical prevalence of the fixed high-D^* bin reported so a rare regime is never mistaken for a vanished wall.

Under the realistic prior with clean acquisition (Arm A), recalibrating *within* the distribution restores marginal coverage to nominal ($D/D^*/f = 0.903\,_{[0.896,\ 0.913]},\ 0.900\,_{[0.884,\ 0.917]}$, $0.898\,_{[0.885,\ 0.912]}$), while naive transfer of the uniform-calibrated predictor breaks (D^* marginal $0.767\,_{[0.735,\ 0.805]}$). The high-$D^*$ wall persists after recalibration in *both* stratifications – fixed-edge tercile coverage $0.533\,_{[0.480,\ 0.594]}$ and per-distribution $0.786\,_{[0.749,\ 0.823]}$, against nominal 0.90 – with the fixed-edge CRLB-to-bin-width ratio reaching $0.895\,_{[0.849,\ 0.961]}$ at high D^* (Fig. 10A,B). As in the shrinkage dissociation (Sec. 4.9), a sub-unity CRLB-to-width ratio coexisting with a large conditional-coverage gap is expected, not contradictory: the ratio bounds *point* precision, whereas the gap is a *conditional-coverage* failure on a latent axis. The severest under-coverage concentrates in a clinically *rare* high-D^* tail (fixed high-D^* prevalence $0.049\,_{[0.044,\ 0.056]}$), which entangles two effects: the rarity of the bin leaves the recalibration split few high-D^* examples for the single global conformal offset, and the identifiability wall itself. A calibration-size control separates them. Up-sampling the fixed high-D^* *calibration* count to the uniform-prior level ($\sim 73 \to 504$ voxels), drawn from the realistic generator at fixed within-bin distribution – a fidelity gate confirms the up-sampled voxels match the realistic, not the uniform, high-D^* law – with train, test and prior held fixed, lifts fixed-edge high-D^* coverage from 0.533 only to $0.661\,_{[0.580,\ 0.721]}$, closing $\approx 47\%$ of the way to the uniform-prior anchor $0.807\,_{[0.774,\ 0.831]}$ and never reaching it (count-matched coverage stays ≤ 0.72 in every seed; Fig. 10D). Rare-bin calibration sparsity thus accounts for roughly half of the fixed-edge deficit; the remainder is a genuine identifiability wall that survives count-matching – a scoping caveat on *prevalence* and a partial contributor to the fixed-edge number, not an explanation of the wall. That the regime is rare does not make it low-stakes: a high-D^* voxel is often the salient finding (brisk lesion perfusion), so under-coverage concentrated in the rare tail falls exactly where a wide, honest interval matters most. The realistic prior is a within-support covariate shift, so the label-free monitor does not flag it (naive-transfer AUC $0.480\,_{[0.470,\ 0.494]}$, at chance); recalibration, not monitoring, is what restores coverage.

Holding the deployed predictor fixed and sweeping the nuisance magnitude (Arm B), marginal coverage degrades monotonically: D coverage falls from $0.903\,_{[0.889,\ 0.912]}$ at $\eta = 0$ to $0.570\,_{[0.548,\ 0.589]}$ at $\eta = 1$, while the already-wide D^* interval stays near nominal ($0.899\,_{[0.888,\ 0.911]}$), and recalibrating within the nuisance-laden distribution restores it (D^* $0.899\,_{[0.886,\ 0.914]}$ at $\eta = 1$). This D^* stability reflects interval *width*, not robust identification: the high-D^* band is already wide enough to absorb the added nuisance, so flat D^* coverage is the characterized-but-unresolved wall showing through, not evidence that D^* is well determined under nuisance. Notably, this acquisition nuisance is *not* more observable than pure signal-model drift: the monitor's AUC stays at chance across the sweep ($0.493\,_{[0.482,\ 0.507]}$ at $\eta = 1$, cf. the off-model envelope's ≤ 0.53), so a measurement nuisance that substantially breaks coverage is a largely *silent* channel – again locating the operative safeguard in recalibration rather than the observable monitor.

5 Discussion

The thread connecting these results is an *observable-vs-latent* distinction. Conformal prediction guarantees *marginal* coverage and, with weighting or stratification on *observable* quantities (SNR, signal features, estimated density ratios), can recover coverage conditional on those observables. What it cannot label-free-guarantee is coverage conditional on a *latent* axis that the data do not identify. In IVIM that latent axis is the true D^* in its ill-posed regime: the Fisher information there collapses, the CRLB exceeds the bin width, and no observable proxy – including the plug-in $\hat{D}^*$ –

recovers the regime. This is why an observable deployment monitor catches every *distribution* shift but is blind to the *within-distribution* high-D^* gap. Four independent stresses corroborate that this split is information, not an artifact of our estimator or generator: a deliberately biased shrinkage prior buys D^* point precision but not high-D^* conditional coverage (Sec. 4.9); the gap replicates when the ground truth is a non-bi-exponential velocity-dispersion process, a forward-model break that is itself a hidden channel the monitor cannot see (Sec. 4.10); the off-model validity envelope degrades gracefully with a recovered bi-exponential origin (Sec. 4.11); and the wall persists under a clinically realistic correlated, skewed prior with measurement-level nuisance, where it concentrates in a clinically rare high-D^* tail and the nuisance proves a silent channel the monitor does not see (Sec. 4.12).

The same wall appears in adjacent label-free problems: deployment-validity monitoring is provably blind to shifts that leave observable statistics unchanged (a "hidden channel"), and the absence of in-vivo ground truth (the "Echo tension") forecloses any direct coverage check on real data. We read these not as failures of conformal prediction – whose guarantee is exactly as advertised – but as the IVIM instance of a known boundary [1]: distribution-free tools deliver everything the data identify and nothing they do not. On what this delivers, stated plainly: the finite-sample coverage guarantee is a property of the synthetic calibration and transfers in vivo only under an exchangeability assumption that cannot be verified where no ground truth exists – it is not an in-vivo coverage claim. What the method adds over the existing IVIM repeatability literature is mechanistic and operational, not a usable per-voxel in-vivo interval: (i) conformalizing a strong model-based base gives the sharpest valid intervals on synthetic data; (ii) an observable monitor guards exchangeability at deployment, while remaining blind to the latent high-D^* gap by construction; and (iii) the high-D^* compartment is reported as characterized-but-unresolved, with a per-voxel reason – the identifiability wall – rather than a trusted point map.

6 Limitations

Our coverage results are *synthetic-primary* by necessity: conformal calibration needs labels that in-vivo IVIM lacks, so the guarantee is established on a forward model we control, and the in-vivo component is explicitly qualitative with no coverage claim. We stress that this is a structural impossibility, not a study limitation removable with more data: because in-vivo IVIM admits no ground-truth (D, D^*, f), realized coverage is unmeasurable in vivo in principle, so a distribution-free coverage guarantee can only ever be synthetic-established and then transported under an explicitly monitored exchangeability assumption – precisely what the real-data section (Sec. 4.7) exercises, rather than a coverage claim it could never support. The base synthetic cohort uses uniform priors over published ranges and Rician noise; real tissue is more heterogeneous. We address this directly with a capstone realism cohort (Sec. 4.12) that draws a correlated, right-skewed joint prior with a published-range D^* coefficient of variation and adds measurement-level nuisance (partial-volume mixing, bulk-motion phase, non-Rician multi-coil noise): the marginal guarantee returns on recalibration and the high-D^* wall persists, concentrated in a clinically rare high-D^* tail. The off-model behavior is likewise characterized rather than probed at a single point: the high-D^* wall replicates under a non-bi-exponential velocity-dispersion generator (Sec. 4.10) and we map a validity envelope across three deviation families with a monitored, graceful degradation and a recovered bi-exponential origin (Sec. 4.11). These remain schematic deviations and a still-synthetic cohort spanning the neighbourhood of the bi-exponential model, not a validated tissue model, and the in-vivo unverifiability above is unchanged by them. The CRLB diagnostic uses the high-SNR Gaussian approximation to Rician noise; the qualitative under-identification of D^*

is noise-model-independent (the D^* Jacobian column vanishes as D^* grows), but absolute bound values at SNR $\lesssim 10$ would shift under an exact Rician treatment. Finally, all coverage guarantees assume exchangeability of calibration and deployment data; the monitor quantifies, but cannot by itself repair, departures from it.

7 Consistency check (reproducibility)

Every numeric claim in this manuscript is drawn from a gated, seeded checkpoint printout. The script `gauge/paper/consistency.py` enforces two contracts: (i) every finite-sample-guaranteed marginal value and every determinism-gated quantity appears *verbatim* in its seed-`20260613` report (`results/*.txt`, `POSITIONING.md`); and (ii) every conditional / empirical (E) headline ships as the *across-seed mean* over a 16-seed re-run and is asserted to lie within its empirical $[5, 95]$ band, which carries an `n_seeds` (`results/multiseed.json`, `multiseed_report.txt`; old→new point map in `results/ci_rebaseline_table.txt`). The full pipeline regenerates deterministically from seed `20260613` (`python -m gauge.baselines`; `gauge.benchmark`; `gauge.conditional`; `gauge.conditional_attack`; `gauge.robustness`; `gauge.invivo`; `gauge.dissociation`; `gauge.altmodel`; `gauge.osipi`; `gauge.realism`), with the core bands from the multi-seed driver (`python -m gauge.multiseed`) and the external-phantom, alternative-forward-model, and realism bands from their own seed sweeps (`python -m gauge.osipi sweep 16`; `gauge.altmodel sweep 16`; `gauge.realism sweep 16`); the realism run also emits the NF1 calibration-size control and the verbatim prior-provenance table. Seed-0 outputs are byte-identical across independent full re-runs.

8 Conclusion

This work brings distribution-free conformal coverage to the ill-posed IVIM inverse problem and benchmarks it against model-based UQ. The naive hypothesis – that the conformal advantage concentrates marginally in D^*/f – is rejected: among credible model-based baselines the marginal overconfidence is mild and broad rather than D^*-specific, conformal restores coverage everywhere, and conformalizing the MDN is the sharpest valid recipe. The genuine IVIM-specific phenomenon is conditional: the high-D^* regime under-covers for every label-free method we tested – including a deliberately biased shrinkage estimator – and the gap replicates under a non-bi-exponential generator. This is the IVIM instance of the known impossibility of distribution-free conditional coverage [1]: the conditioning axis, the true D^* regime, is latent and unidentifiable. The Cramér–Rao bound, which exceeds the regime resolution here, is a concomitant point-identifiability diagnostic, not the demonstrated cause. We therefore characterize the limit – and provide the robustness tooling, acquisition analysis, and qualitative in-vivo demonstration that frame how an honest, distribution-free IVIM uncertainty map should be deployed and monitored.

Data and Code Availability Statement

All coverage results are produced from a deterministic, seeded synthetic cohort (seed `20260613`) generated by the study's own forward model; no external or in-vivo dataset is required to reproduce them. The complete source code, the gated checkpoint outputs from which every number in this paper is drawn, the figures, and a machine-checked consistency script (`gauge/paper/consistency.py`) are openly available in the project repository and permanently archived at https://doi.org/10.5281/zenodo.20686273 (the all-versions DOI, which resolves to the latest archived release). The qualitative in-vivo cross-check (Sec. 4.7) applies the as-deployed predictor and monitor to real multi-b DWI from the publicly available ACRIN-6698 / I-SPY2 Breast DWI collection (The Cancer Imaging Archive; DOI `10.7937/tcia.kk02-6d95`; licensed CC-BY-4.0) [26, 27], which has no ground-truth IVIM parameters and supports only qualitative pipeline/monitor behavior and a label-free same-day test–retest repeatability-tracking check (n=76 tumors, Sec. 4.7)—it makes no in-vivo coverage claim. Under a download-on-demand posture no pixel data are redistributed here; the data are retrieved by `scripts/fetch_invivo.py` into an ignored path, and only the provenance manifest (`results/invivo_real_provenance.json`, with series identifiers, b-values, license, and citation) is committed. The ACRIN-6698 imaging is publicly available and de-identified, requiring no additional ethical approval for this secondary, qualitative analysis; the synthetic coverage pipeline uses no human or animal data and regenerates independently of any external dataset. The external-phantom replication (Sec. 4.8) uses the OSIPI TF2.4 IVIM reference digital reference object—an independent *synthetic* ground truth (code Apache-2.0; data Zenodo `doi:10.5281/zenodo.14605039`, CC-BY-4.0) [28], retrieved on demand by `scripts/fetch_osipi.py` with only the provenance manifest (`results/osipi_provenance.json`, with record DOI, checksums, units, and citation) committed; it is a synthetic reference, not in vivo, and makes no in-vivo coverage claim.

Conflict of Interest

The author declares no competing interests.

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Author Contributions

Avery Karlin is the sole author and conceived the study, designed and implemented the methods and software, performed the analysis, produced the figures, and wrote the manuscript.

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Figure 7: Real in-vivo data (ACRIN-6698 / I-SPY2 Breast DWI; qualitative, no coverage claim). (Top left) the re-fit conformal D^* band vs the plug-in $\hat{D}^*$; (top right) the D^* band-width distribution and the as-deployed monitor verdict (fires, AUC 0.97, via the b-independent Mahalanobis family). (Bottom) test–retest: the conformal D-width vs the measured scan–rescan ADC variability across n=76 same-day-repeat tumors (Spearman $r = +0.60$, 95% CI $[0.42, 0.72]$; robust, repeatability-tracking only—not validation or coverage).

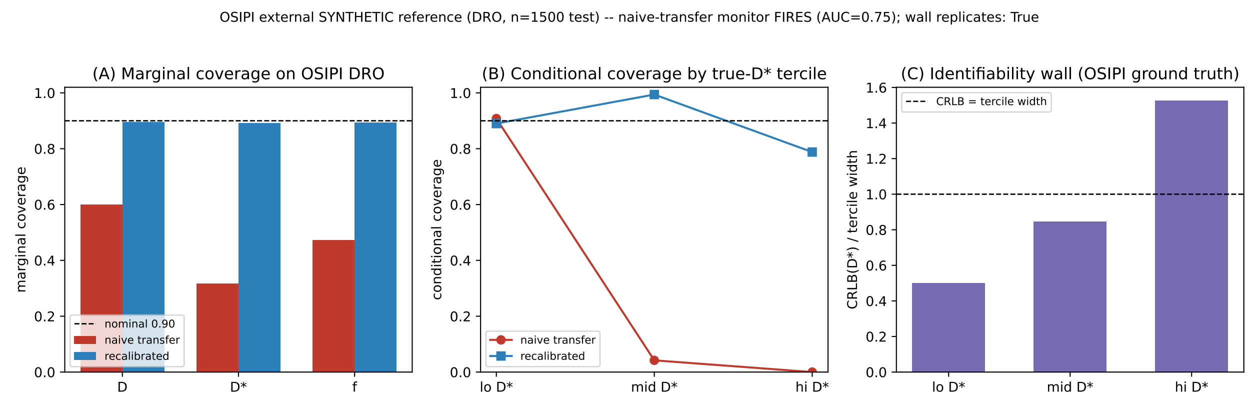

Figure 8: OSIPI external *synthetic* reference (DRO; not in vivo, no in-vivo coverage claim). (A) Marginal coverage vs nominal: naive transfer of our synthetic calibration breaks, recalibrating within the OSIPI distribution restores it. (B) Coverage conditional on the true D^* tercile: the high-D^* gap persists even after recalibration. (C) The identifiability wall on OSIPI ground truth – CRLB(D^*) exceeds the high-D^* tercile width. The naive-transfer monitor fires (AUC 0.74), so the break is flagged.

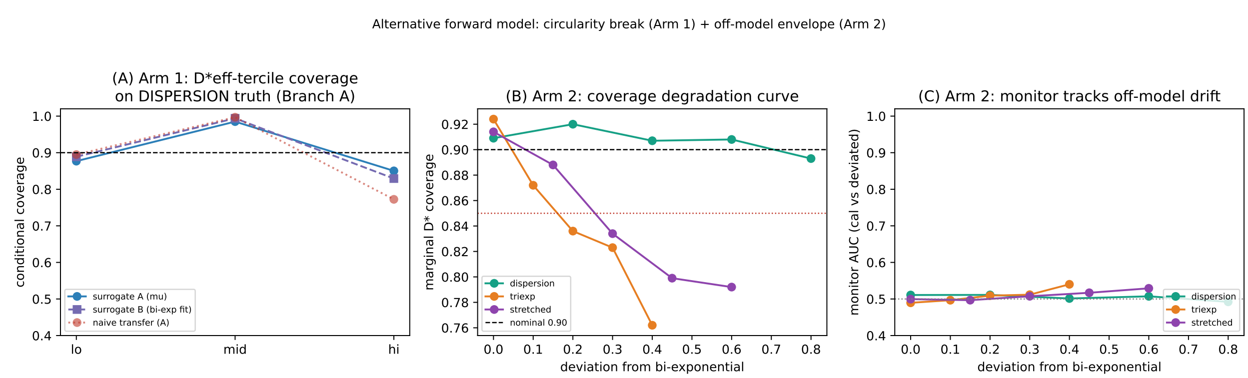

Figure 9: Alternative forward model. (A) Arm 1 – D^*_{eff}-tercile conditional coverage when the ground truth is a non-bi-exponential velocity-dispersion process: after within-distribution recalibration the high-D^*_{eff} gap persists on both surrogate axes (primary gamma mean A, corroborating bi-exp-fit B), while naive transfer breaks. (B) Arm 2 – marginal D^* coverage degrades monotonically as the forward model deviates from bi-exponential across three families, recovering to nominal at zero deviation. (C) Arm 2 – the deployment monitor's AUC vs. deviation: discrimination of off-model drift is weak and family-dependent. Across-seed means (16 seeds); from `results/altmodel_report.txt`.

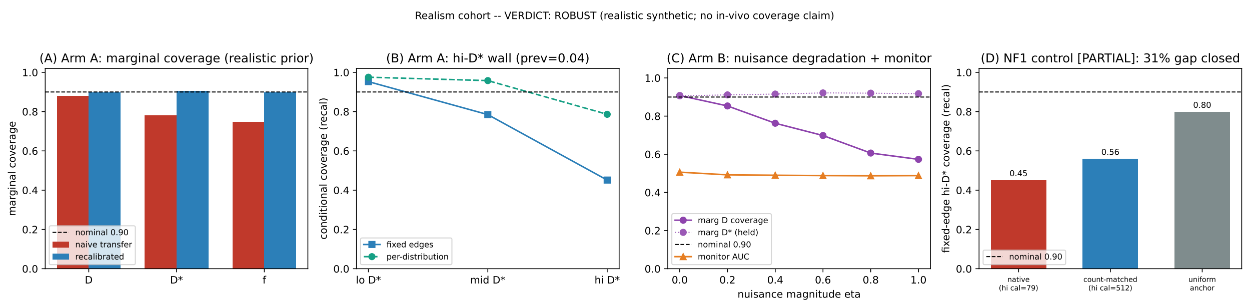

Figure 10: Realism cohort (realistic synthetic; no in-vivo coverage claim). (A) Arm A – marginal coverage under a realistic correlated/skewed prior: naive transfer of the uniform-calibrated predictor breaks, within-distribution recalibration restores nominal coverage. (B) Arm A – recalibrated high-D^* tercile coverage on fixed physical boundaries vs. per-distribution quantiles; the wall persists in both, concentrated in a clinically rare high-D^* tail (fixed prevalence ≈ 0.05). (C) Arm B – the measurement-nuisance envelope: marginal D coverage degrades with the nuisance magnitude (the wider D^* interval is held) while the deployment monitor's AUC stays at chance, a silent channel. (D) NF1 calibration-size control: fixed-edge high-D^* recalibrated coverage under native vs. count-matched calibration (hi-D^* calibration count $73 \to 504$), against the uniform-prior anchor; only $\approx 47\%$ of the native$\to$uniform gap is closed by count-matching, isolating a residual wall. Seed-`20260613` panel (across-seed [5,95] bands quoted in text); from `results/realism_report.txt`.

Supporting Information

The following table and figures support the main text and are provided as Supporting Information.

Table 7: Realistic joint-prior specification (§4.12, Arm A and the NF1 calibration-size control). Hyperparameters are documented design constants *shaped to approximate* representative abdominal IVIM patient cohorts [9, 11, 12], not fitted; the joint is a Gaussian copula on latent normals with the marginals below. Committed verbatim in `results/realism_report.txt` and `results/realism_provenance.json`.

Parameter	Marginal	Hyperparameters
D	log-normal	mean 1.1×10^{-3} mm^2/s, CV 0.30
D^*	log-normal	mean 28.0×10^{-3} mm^2/s, CV 0.80 (target band 0.50–1.10)
f	logit-normal	logit-mean ln(0.20/0.80), logit-sd 0.55 (median $f \approx 0.20$)
SNR(b=0)	log-normal	mean 35.0, CV 0.40, clipped to $[8.0, 120.0]$
latent-normal correlation $(D, D^*, f) = \begin{pmatrix} 1.00 & 0.00 & -0.20 \\ 0.00 & 1.00 & 0.35 \\ -0.20 & 0.35 & 1.00 \end{pmatrix}$		

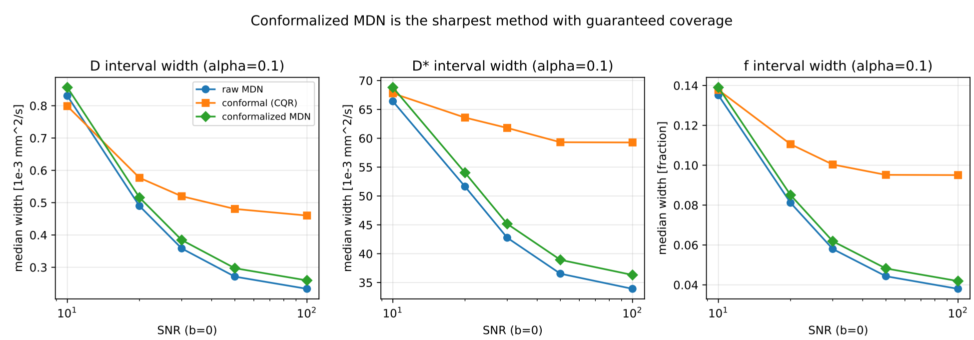

Figure S1: Median interval width vs SNR. Conformalized-MDN is the sharpest method with a guaranteed coverage level.

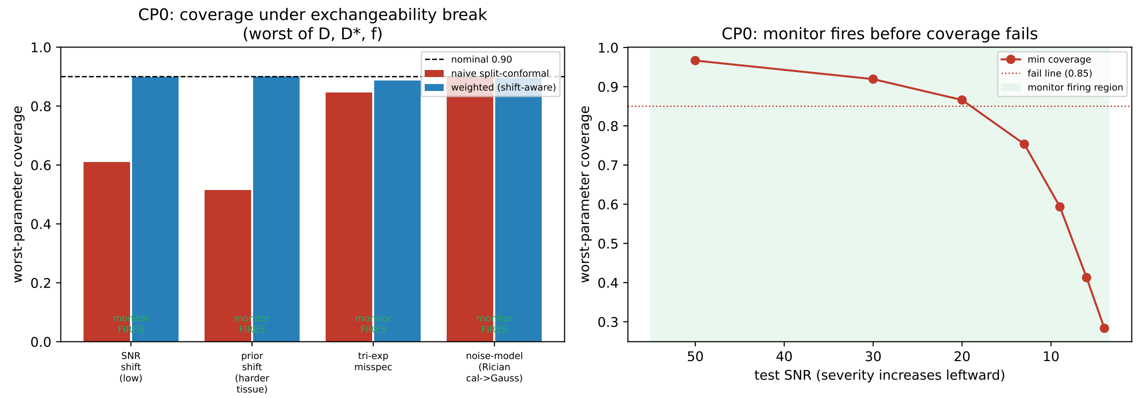

Figure S2: (Left) worst-parameter coverage under each exchangeability break, naive vs. weighted, with monitor status. (Right) SNR-severity sweep: the monitor fires before coverage crosses the failure line.

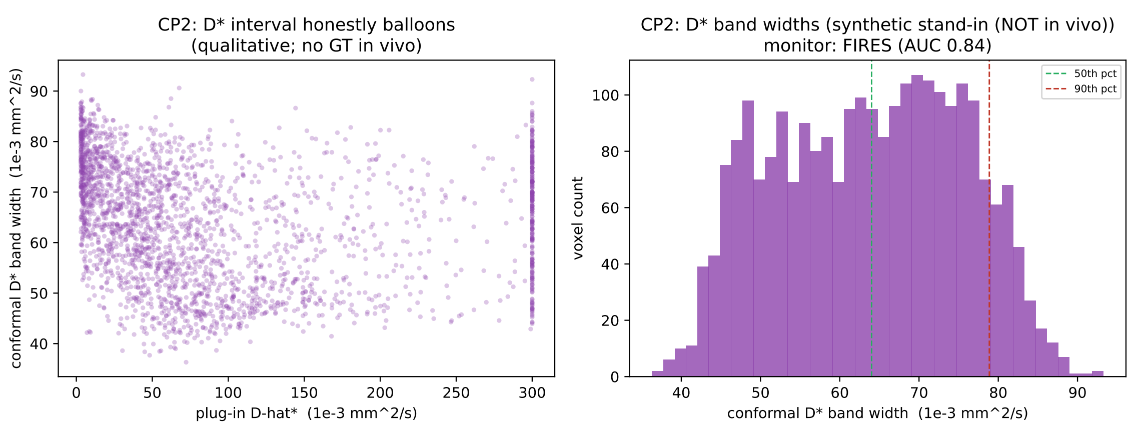

Figure S3: In-vivo qualitative demo (synthetic stand-in). (Left) the conformal D^* band balloons with the plug-in $\hat{D}^*$. (Right) the heavy-tailed D^* band-width distribution; the deployment monitor fires on the transfer.